

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

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The Theory of Financial Crises (Chapter 8)



Outline of class

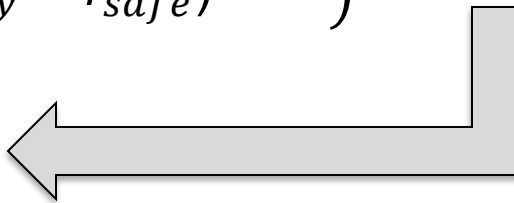
✓ General theory of Financial crises

1. Definition
2. Origins of Financial Crises: 3 stages
3. Some major episodes: Japan, 1929 and Financial Crisis

Only Financial crises in “advanced” economies

Definition

- ✓ **Financial crises** are major disruptions in financial markets characterized by
 - sharp declines in asset prices
 - failures of Financial Institutions
 - some government policy intervention afterwards
- ✓ **Financial crises** occur when information flows in Financial Markets (FMs) experience a very large disruption
 - financial frictions (read Asy infos in FMs)
and
credit spreads ($= r_{risky} - r_{safe}$)
} increase sharply
 - FMs stop functioning!



Definition

Financial crises $\neq$ Recessions

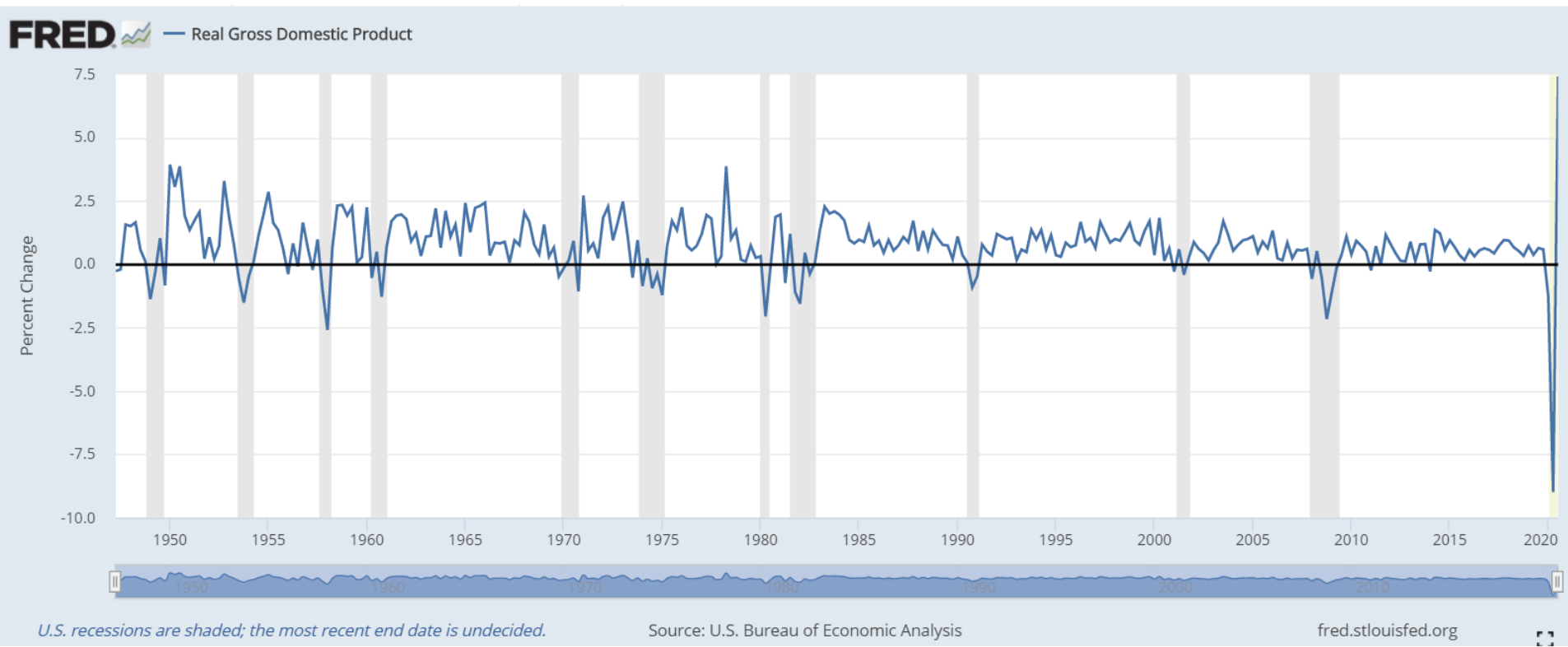
- ✓ A **recession** indicates an economic contraction (3-4 consecutive quarters with $\Delta\text{GDP} < 0$).
- ✓ A financial crisis **causes** a recession
- ✓ But recession **not just caused by** financial crises

Definition

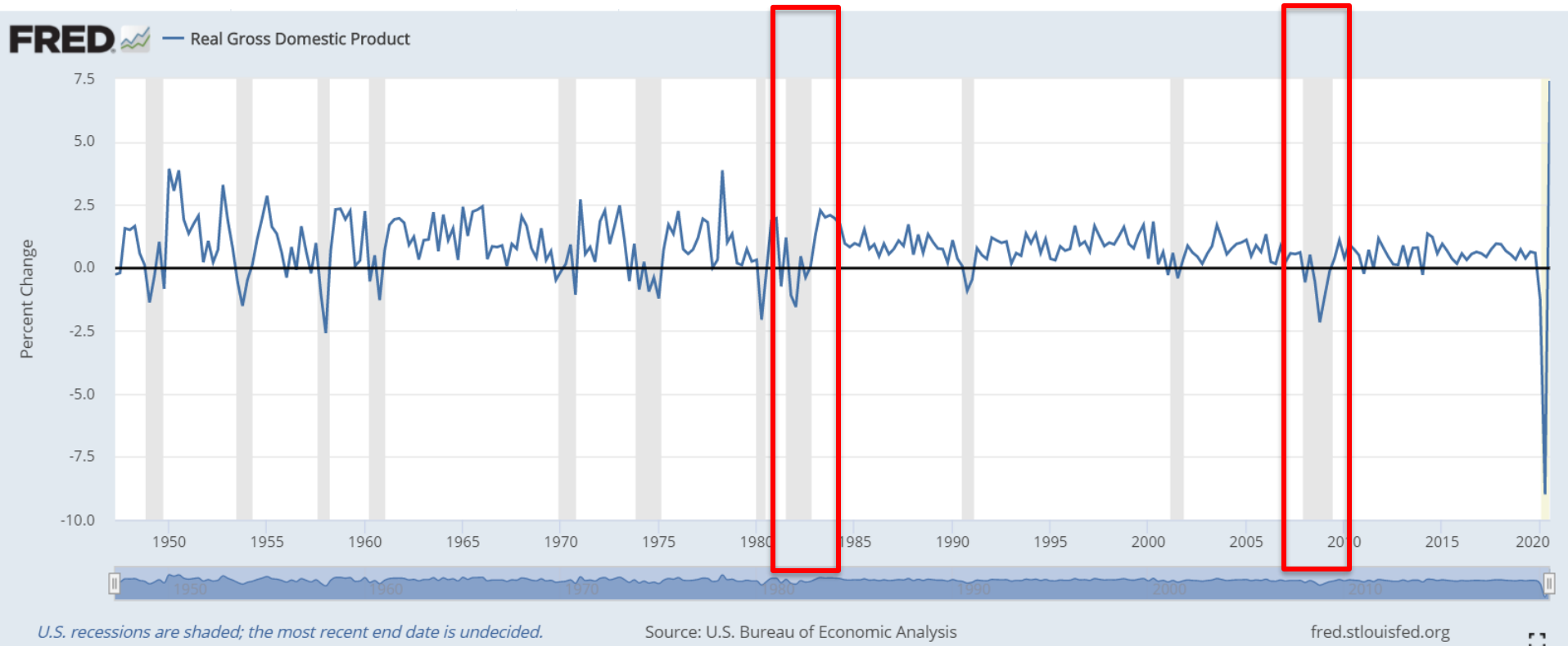
Financial crises $\neq$ Recessions

- ✓ A **recession** indicates an economic contraction (3-4 consecutive quarters with $\Delta\text{GDP} < 0$).
- ✓ A financial crisis **causes** a recession
- ✓ But recession **not just caused by** financial crises
- ✓ Recessions can be caused **also by**:
 - a *supply* shock (the 1973 oil crisis, 2022 energy crisis)
 - a *trade/political* shock (collapse of the USSR in Dec 1991)
 - a natural *disaster* (earthquake and tsunami in Japan 2011)
 - a *pandemic* (Covid-19 in 2020)

US Recessions and Financial Crises



US Recessions and Financial Crises

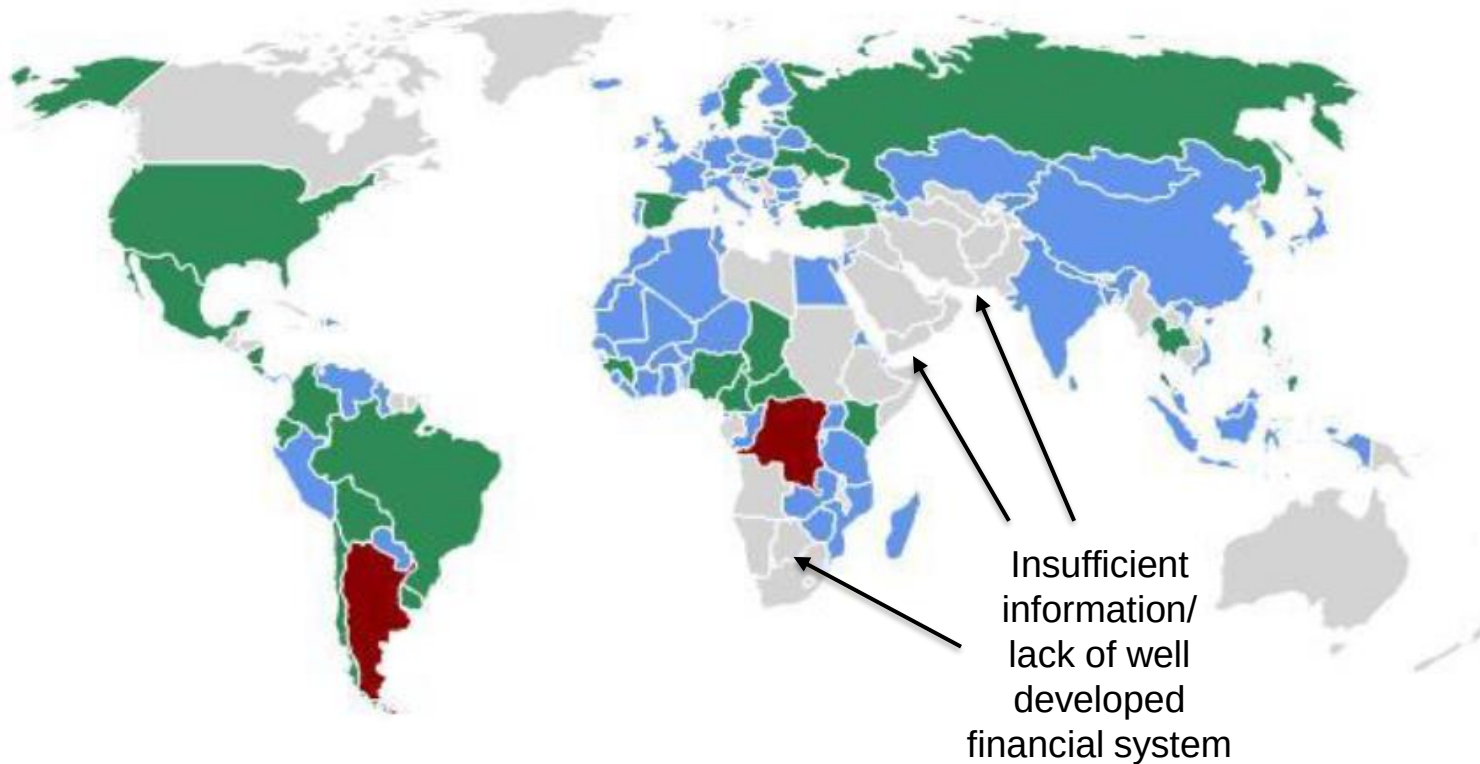


Financial/Banking Crises

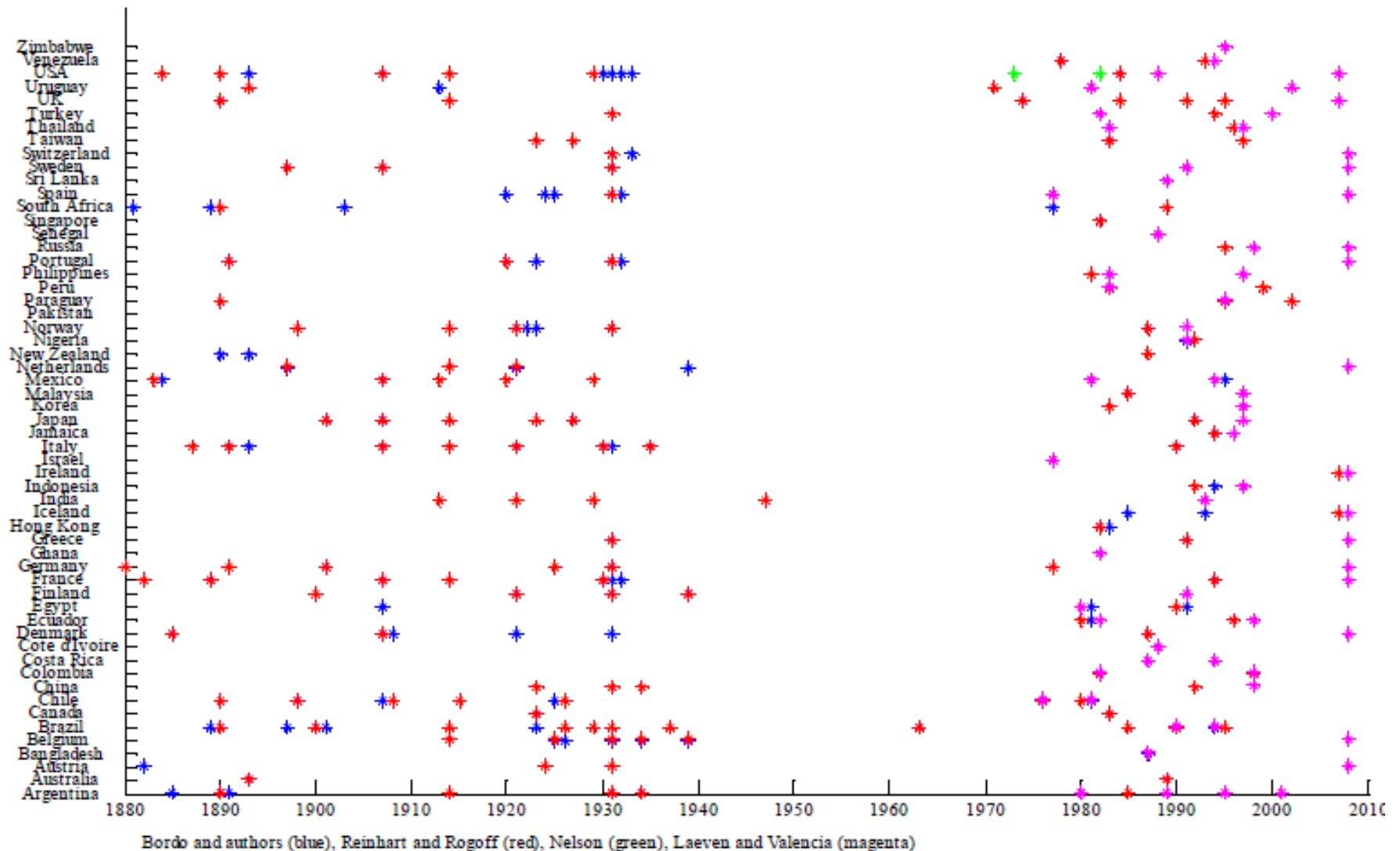
Financial Cris*ES*

- ✓ Financial Crises are a prevalent feature of our (i.e. capitalist) economies
- ✓ Lower frequency event compared to recessions, but ...
- ✓ ... more frequent in the past:
 - general frequency has been high
 - 15 banking/financial panic between 1793-1933 in the US
 - only 2 after 1933
 - Bretton-Woods period (1944-1973 is a big exception)
- ✓ Over the last 120 years, financial crises worldwide have
 - lasted *2-3 years*
 - and **cost 5-10% of GDP** on average
 - struck almost everywhere (look at the map: next slide)

Banking Crisis Worldwide 1970-2011



Banking Crisis Worldwide 1880-2010



Bordo and authors (blue), Reinhart and Rogoff (red), Nelson (green), Laeven and Valencia (magenta)

Why study Financial Crises?

- ✓ Since history tends to repeat itself ...
 - ... by studying financial crises we can learn from past mistakes
 - and try to prevent/contain *future* crises
 - [Queen Elizabeth II](#)

- ✓ For example:
 - Ben Bernanke was appointed in 2006 as Fed chairman
 - academic from Princeton and scholar of the Great Depression
 - his research (2022 Nobel prize) told him that the Fed did not appropriately respond to the 1930s crisis
 - the strong Fed response in 2008 and afterward probably averted a much more prolonged recession in the US

 - ... what about Alan Greenspan?
 - he was chairman of Fed from 1987 to 2006

Origins of Financial Crises: Stages

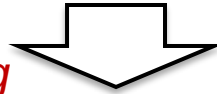
Stage 1: **Initiation (Debt Build-up and deleverage)**

A. credit boom

B. bubble

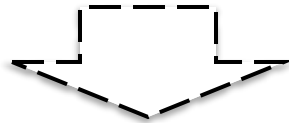
C. increase in uncertainty

deleveraging



Stage 2: **Banking crisis (Real Economy)**

○ *fire sales*



Stage 3 **Debt deflation** (does not always happen)

Origins of Financial Crises

- ✓ Financial Crises seem to have a similar pattern
- ✓ They are typically preceded (in good times) by
 - **credit booms**
 - and
 - **asset price bubbles**
- ✓ credit booms:
 - every single financial crisis happened **because of excessive debt**, one way or another
 - many of those booms happened in real estate
- ✓ The seeds are often planted with ...
 - ... *financial* **liberalization**
 - ... or *financial* **innovation**

Stage 1: Initiation

A. Credit Boom (and then bust)

B. Bubble: asset-price boom (and then bust)

- often accompanies a credit boom&bust

C. Increase in Uncertainty

- caused by:
 - the *failure of some major FI*
 - Ex.: Lehman and AIG in 2008, Knickerbocker Trust in 1907
 - start of a recession
 - crash of stock market (Walls Street 1929)
- balance sheets of FI's deteriorate, FI's start to **deleverage**

Stage 1: Initiation

A. Credit boom

- aggregate amount of credit in the economy is higher than its long-run trend... in other words, banks lend a lot – maybe “too much” ([read](#))
- ✓ Additional credit increases demand for some assets, which appreciate substantially in price (**asset price boom**)
 - price increase of some asset used as *collateral* tends to boost lending even more!
- ✓ Often credit booms follow a financial **liberalization** ...
 - regulation allows *more competition* between banks
 - the country liberalizes *international capital flows*
 - the country joins a *monetary union* (ex.: Spain)
- ✓ ... or financial **innovation**
 - introduction of new types of loans or other financial products

Stage 1: Initiation

- ✓ It's **not** that **all credit booms lead to a recession** ...
... otherwise we'd better off without finance!
 - often credit booms are simple expansions of credit with no follow-up problems, and are good for economic development

- ✓ An IMF study found that in 135 credit booms across 100 countries
 - **104** have a “good” end (good)
 - **31** are followed by a banking distress... (bad)
 - ...of which **23** are systemic crises (ugly)
 - ... so have a probability of **17%** that a credit boom becomes a disaster
 - high, but still *positive* effect most of the time

Stage 1: Initiation

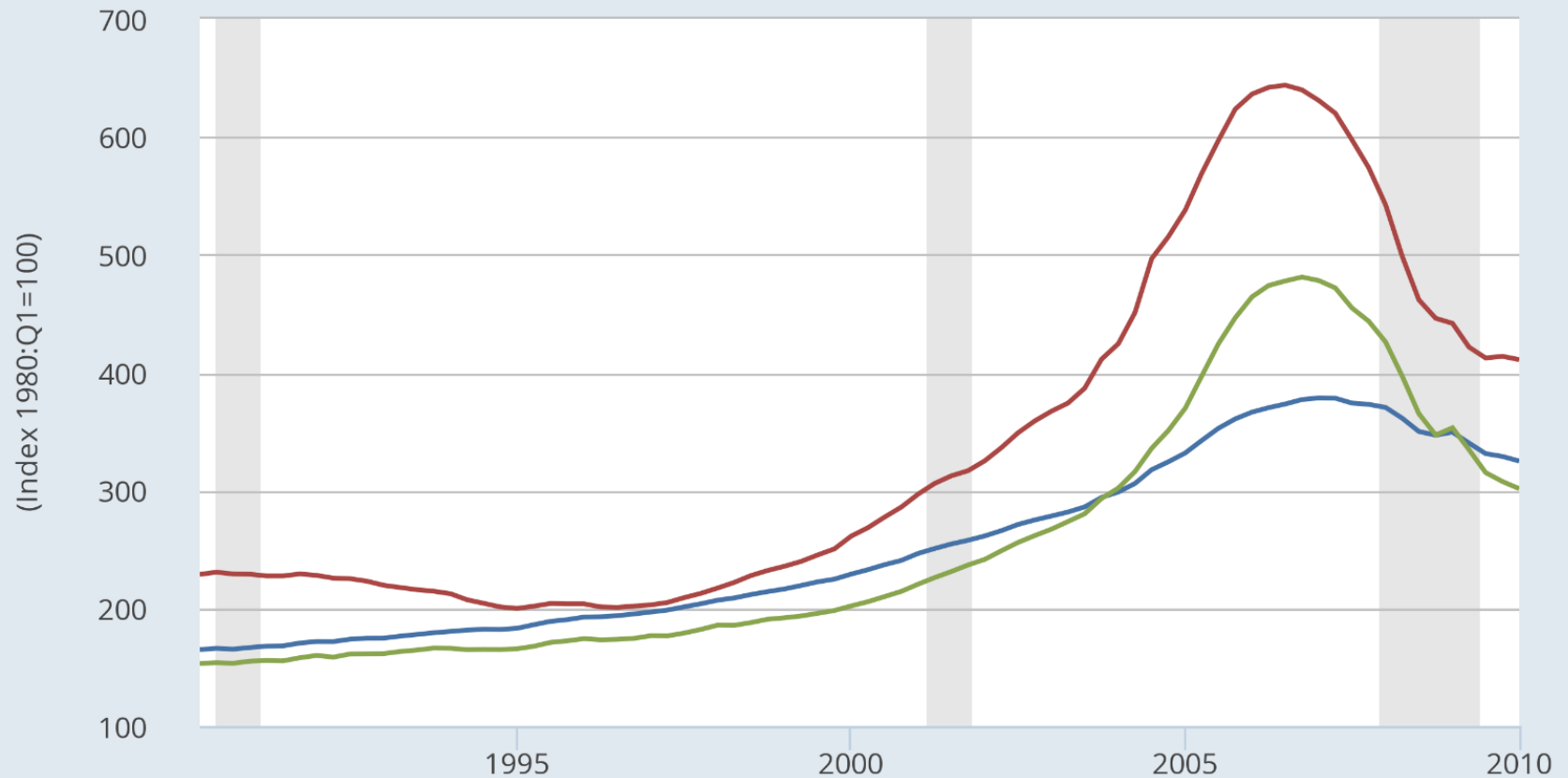
B. Bubble

- the price of an asset is well above the **fundamental value** (=PV of future cash flows)
- ✓ « $P > \text{fundamentals}$ » means:
 - an “irrealistic” price, unjustified by current conditions or predicted earnings
- ✓ we saw this can happen with stocks fairly easily
 - ... because of overconfident investors
 - nice start reading on bubbly assets: [review paper](#) (optional)
- ✓ but it can also happen with other assets
 - one especially: housing (*real estate*)

Credit Boom: Housing Bubble

FRED 

- All-Transactions House Price Index for the United States
- All-Transactions House Price Index for California
- All-Transactions House Price Index for Florida

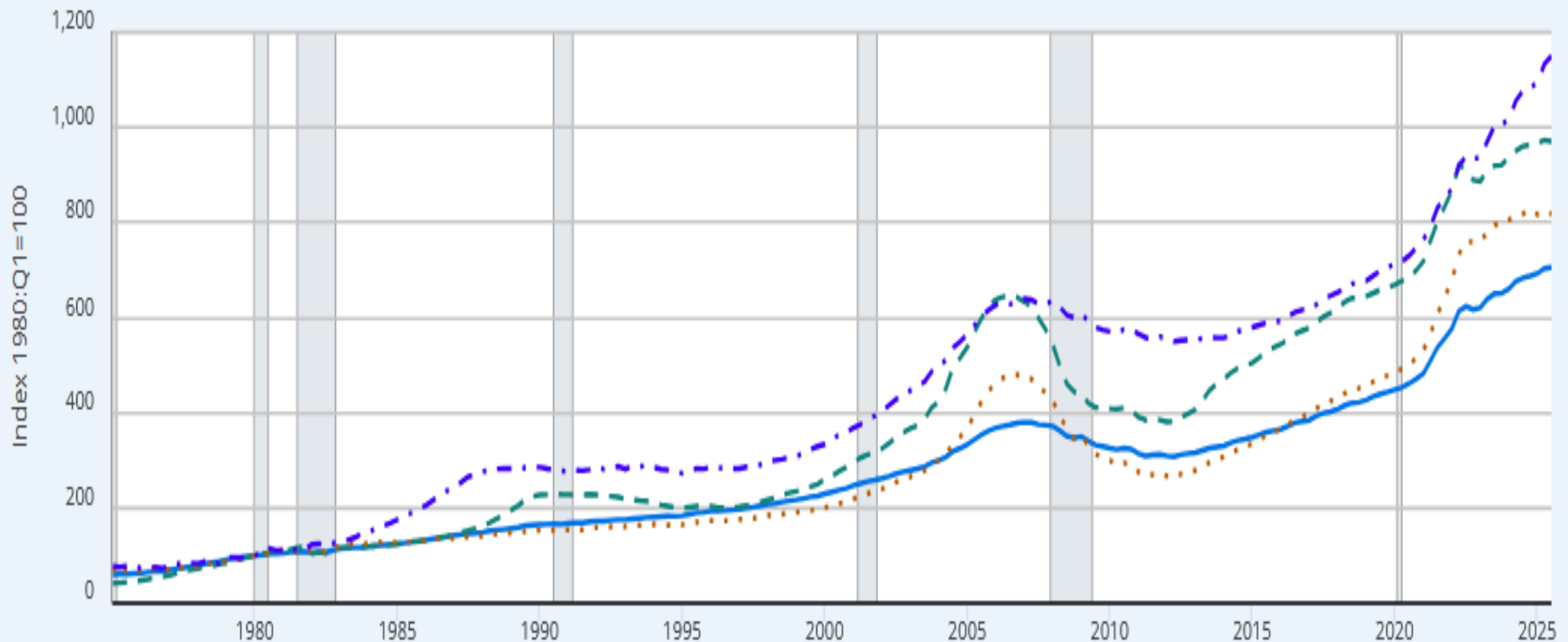


Credit Boom: Housing Bubble

FRED

- All-Transactions House Price Index for the United States
- - All-Transactions House Price Index for California
- ... All-Transactions House Price Index for Florida
- · - All-Transactions House Price Index for New York

where do we stand now?



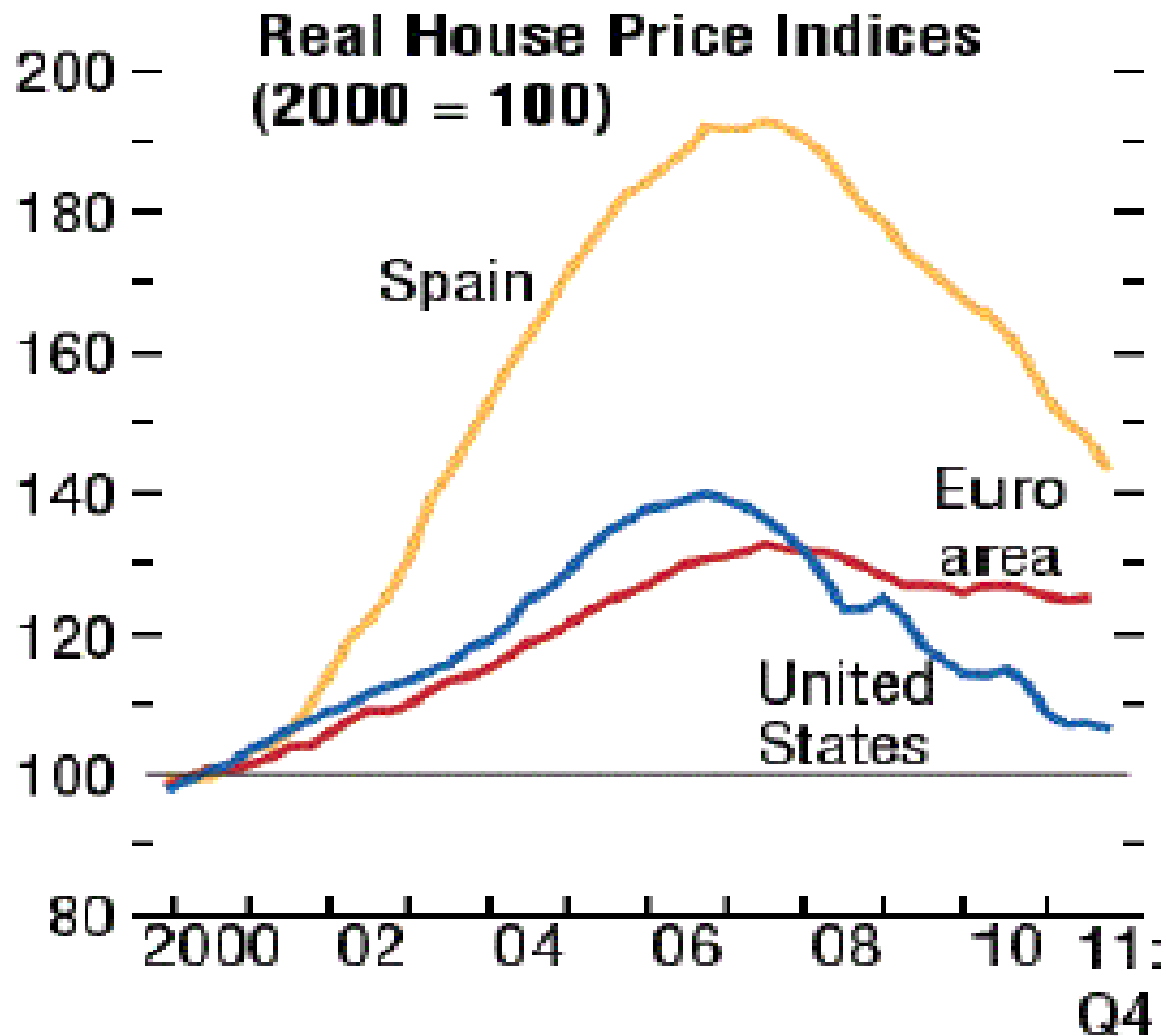
Source: U.S. Federal Housing Finance Agency via FRED®

Shaded areas indicate U.S. recessions.

fred.stlouisfed.org

Fullscreen

Housing Bubble



Stage 1: Initiation

C. Increase in Uncertainty (*Lending boom/Asset bubble ends*)

- at some point, higher uncertainty: things start going bad
- *loan losses accrue* and *asset values fall*: FI's capital ↓
- So Financial Institutions:
 - cut back in lending (*deleveraging*)
 - become *riskier* and some of their creditors pull out their funds, this again prompts banks to cut *back on lending*...
- ✓ As a result, the economy experiences a credit freeze
- ✓ Government intervenes:
 - but safety nets create *moral hazard*
- ✓ Interest rate *hikes* bring *adverse selection* problems
 - higher interest rates cause *drop in value* of cash flows of firms ...
 - ... firms need more ext. finance, but higher int. rates ...
 - ... deter best borrowers from *credit demand* and only risky ones remain in the credit market

Exercise: Deleveraging

- ✓ Consider a bank with the following balance sheet

Assets		Liabilities	
Reserves	\$5	Deposits	\$80
Loans	\$95	Bank Capital	\$20
Total	\$100	Total	\$100

- ✓ \$50 of loans will mature next quarter
- ✓ The bank can either
 - lend the maturing loans again (*roll-over*)
 - or
 - keep the money as reserves
- ✓ The minimum level of bank capital is 10% of loans

- ✓ *Scenario 1*: All borrowers repay

Q: Can the bank lend \$50 again? Is capital sufficient?

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Q: Can the bank lend \$50 again? Is capital sufficient?

A: Yes, it can. The balance sheet would be identical, and capital is $\$20/\$95 = 21\% > 10\%$ **adequate!**

Exercise: Deleveraging

✓ *Scenario 2:* Only \$40 are repaid (\$10 loss). Moreover, the bank expects further losses for \$5 on remaining loans

Q: Can the bank lend \$40 again? Is capital sufficient?

Exercise: Deleveraging

- ✓ **Scenario 2:** Only \$40 are repaid (\$10 loss). Moreover, the bank expects further losses for \$5 on remaining loans

Q: Can the *bank lend \$40 again*? Is capital sufficient?

A1: No. If it lent \$40 again the balance sheet would be

Assets		Liabilities	
Reserves	\$5	Deposits	\$80
Loans	\$80	Bank Capital	\$5
Total	\$85	Total	\$85

- ✓ Capital falls: from \$20 to \$5 because of losses = \$15 (\$10+\$5)
- ✓ Capital ratio: $\$5/\$80 = 6.25\% < 10\%$ **insufficient!**

Exercise: Deleveraging

✓ *Scenario 2*: Only \$40 are repaid (\$10 loss). Moreover, the bank expects further losses for \$5 on remaining loans

Q: Can the bank lend \$40 again? Is capital sufficient?

A2: The bank could instead *convert* the \$40 of maturing loans into reserves and *not renew* the loans

Assets		Liabilities	
Reserves	\$45	Deposits	\$80
Loans	\$40	Bank Capital	\$5
Total	\$85	Total	\$85

○ Capital ratio: $\$5/\$40 = 12.5\% > 10\%$ adequate!

✓ Is \$40 efficient? No! \$30 is. Why?

The Impact of Deleveraging

- ✓ During a financial crisis, the banks will **deleverage** (cut back on lending) rather than **increase capital**
 - look at *Capital Requirement* = $\frac{E}{RWA}$ 
- ✓ Why? Because issuing equity is very costly during a crisis:
 - with low stock prices, FI would do *not* raise enough funds
 - high risk of *losing money* for all shareholders (incumbent and new)
- ✓ With less credit available, firms can't finance their growth
- ✓ As a result, economic activity **declines**!

banks decrease
denominator (e.g.:
loans) rather than
increase numerator
(e.g.: equity)

Transition to Stage **2**: Banking Crisis

- ✓ As banks reduce lending, the economy suffers
- ✓ The decline in economic activity ...
 - ... further deteriorates bank balance sheets (ex.: bad loans increase)
- ✓ More FI's are likely to default
- ✓ This is the **banking crisis** (stage 2)

Stage 2: Banking Crisis

- ✓ If severe enough, these factors lead to a **bank panic/run**
 - if multiple banks fail, banking **panic** may emerge where everyone attempts to take his money out of the banking system ...
 - ...bcs of Asy Info: **nobody knows which banks have good (bad) loans**, depositors will be **suspicious** of every bank
 - Deposit insurance? Not designed to against simultaneous default of many banks
 - so, **running for your money** in such a situation may be **rational!**

- ✓ **Fire sales**
 - as cash balances fall, even still-solvent FIs must sell illiquid assets *quickly*: not easy unless sell cheaply
 - **fire sales**: prices at which assets are sold tend to be much lower (excessive and quick sell-off induce)
 - but fire sales **further deteriorate balance sheets of banks**
 - because the value of remaining assets falls, and net worth **decreases**

Exercise: Fire Sales

- ✓ Consider the following bank balance sheet

Assets		Liabilities	
Reserves	\$10	Deposits	\$80
Loans	\$90	Bank Capital	\$20
Total	\$100	Total	\$100

- ✓ When held until maturity, all loans are worth their book value
- ✓ However, before maturity, loans can be sold only for **75%** of their book value
- ✓ Assume that the bank cannot increase borrowings and there are no reserve or capital requirements
- ✓ **Q:** How much can depositors withdraw before this bank becomes insolvent?

Exercise: Fire Sales

A: clearly the bank can handle \$10 of deposit withdrawals by drawing down on its reserves



Exercise: Fire Sales

A: clearly the bank can handle \$10 of deposit withdrawals by drawing down on its reserves

- the bank can also **sell off loans**, but at 25% discount.

Let $\$y$ be the amount the bank receives by selling loans:

$$\$y = LAS \times 0.75 \Rightarrow LAS = \$y/0.75 \quad \text{where: } LAS = \text{Loan Amount Sold}$$

Exercise: Fire Sales

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- But fire sale means loan losses ... so, capital will decrease too!
- To guarantee solvency, must have enough capital to cover losses:

$$Capital - Losses \geq 0 \Rightarrow \$20 - LAS \cdot 0.25 \geq 0 \Rightarrow \$20 - \left(\frac{\$y}{0.75}\right) \cdot 0.25 \geq 0$$

$$\$y^* \leq \$60$$

- \$y^*\$: “**maximum**” deposit withdrawal the bank can sustain by selling loans



Exercise: Fire Sales

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$$\$y^* \leq \$60$$

- \$y*: “**maximum**” deposit withdrawal the bank can sustain by selling loans
- Hence, with \$y *= \$60 from loans and \$10 from reserves ...
... the bank can at most sustain a deposit outflow of:
\$60 + \$10 = **\$70** before it is insolvent

Stage 3: Debt Deflation

- ✓ *If* the crisis also leads to a sharp decline in overall consumer prices, **debt deflation** can occur
- ✓ A **deflation** is an episode of
 - *declining* prices
and
 - *expecting* more decreases in the *future*
- ✓ Economic agents
 - consumers happy: they have cheaper consumption goods/services, so they postpone purchases
 - lenders happy: real value of debt increases
 - borrowers **not** happy: **debt deflation makes it harder to service debt**
- ✓ This is the worst possible outcome because it prolongs the stagnation for a long time

Stage 3: Debt Deflation

- ✓ Consumers postpone consumption of durable goods (cars, houses etc..) $\Rightarrow$ Consumption $\downarrow \Rightarrow$ GDP $\downarrow$
- ✓ Deflation is bad for finance too:
 - as $\pi^e \downarrow \Rightarrow r \uparrow$ (remember real interest rate: $r = i - \pi^e$)
- ✓ Debt contracts:
 - typically fixed in *nominal* terms, they do not adjust
- ✓ Hence, when prices fall ...
 - ... the *value* of those fixed payments *increases* in real terms (creditors are still promised the same, fixed nominal amount as before)
 - ... overall debt burden for borrower *increases*!

Stage 3: Debt Deflation

- ✓ Deflation: good for lenders, but bad for borrowers
- ✓ The **real value of debt** becomes bigger. Why?
- ✓ Consider a firm balance sheet:
 1. Asset values do **not rise** ($\bar{A}$)
 - actually, they fall in recession: when prices fall, firms need to sell more products to service the same amount of debt payments, which are fixed in nominal terms. At the same time, selling those products is more difficult because of the recession
 2. Real value of liabilities **increases** in real terms ($L \uparrow$)
 3. Real value of net worth **decreases** (recall $\bar{A} = L(\uparrow) + NW$, so $NW \downarrow$)

Example:

- case 1: assets are not deflated
- case 2: assets are deflated

	before deflation	after deflation	
		case 1	case 2
A	100	100	96
L	80	88	88
NW	20	12	8
$\Delta P/P$	-10,0%		

A Full-fledged Financial Crisis: Japan

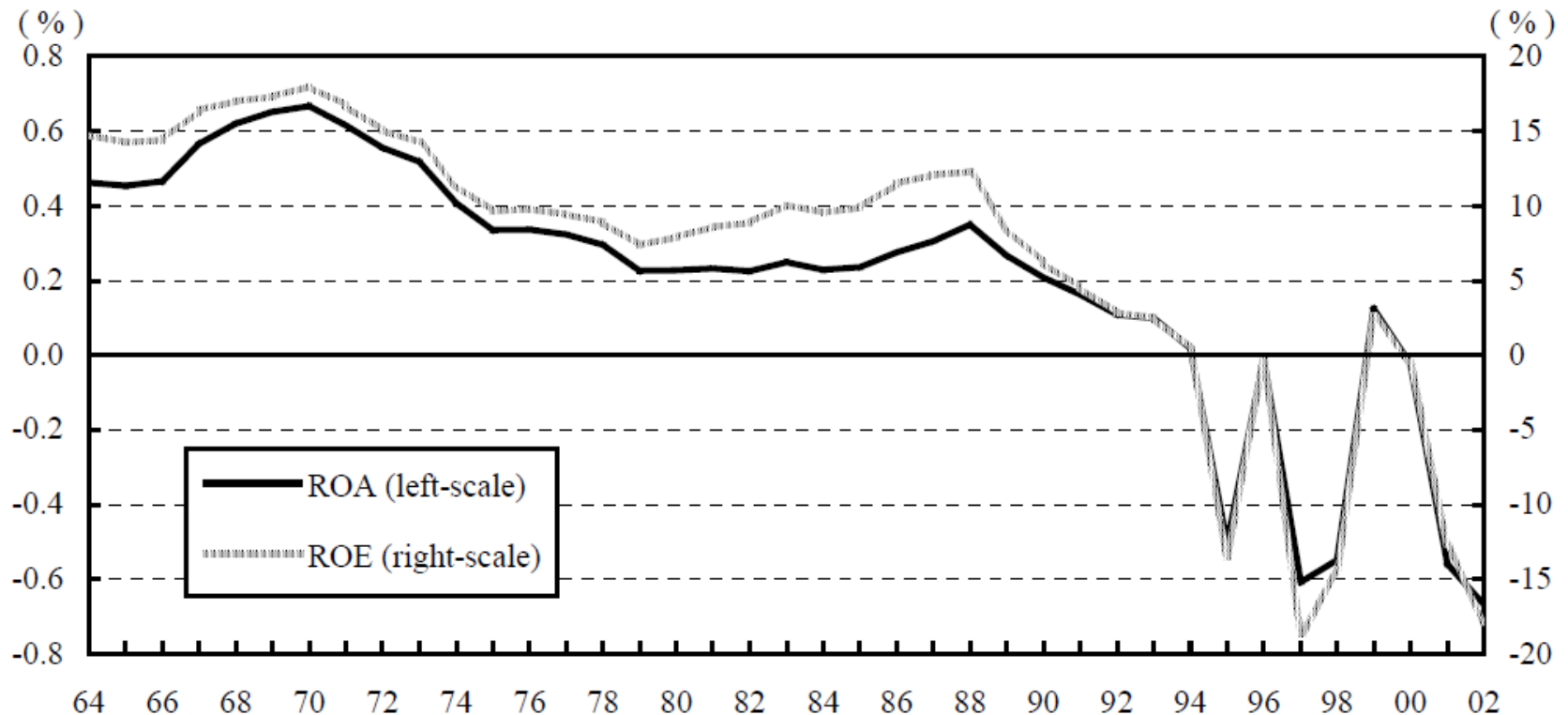
- ✓ Japan in the 1990s experienced a financial crisis that had all the three stages, including a long, prolonged episode of deflation (or rather, missing inflation) and stagnation
- ✓ There are important lessons here, something that policy makers in US and Europe were very keen to avoid when the 2007-09 financial crisis hit
- ✓ Let's have a look at what happened in Japan

Japan before the Recession

- ✓ Japan was an incredible success story of strong economic growth since WWII (like China today)
- ✓ Economy was based on few large **keiretsu**: a conglomerate of companies around a bank who provides the funding
- ✓ **Financial liberalization in 1985**
 - deregulation of *interest rates on deposits*
 - banks paid interest on deposits but did not increase interest on loans (due to competition)
 - profitability of Japanese banks eroded, and losses were partly hidden
 - Basel rules allowed Japanese banks to count unrealized stock gains (up to 45%) for equity
 - Japanese banks borrowed extensively on Euro-dollar markets
 - ¥186tn (\$1.28tn) by June of 1990

Bank profitability in Japan

Profitability of Japanese Banks



Source: Shiratsuka (2003)

Credit Boom fuels Japanese Housing Bubble

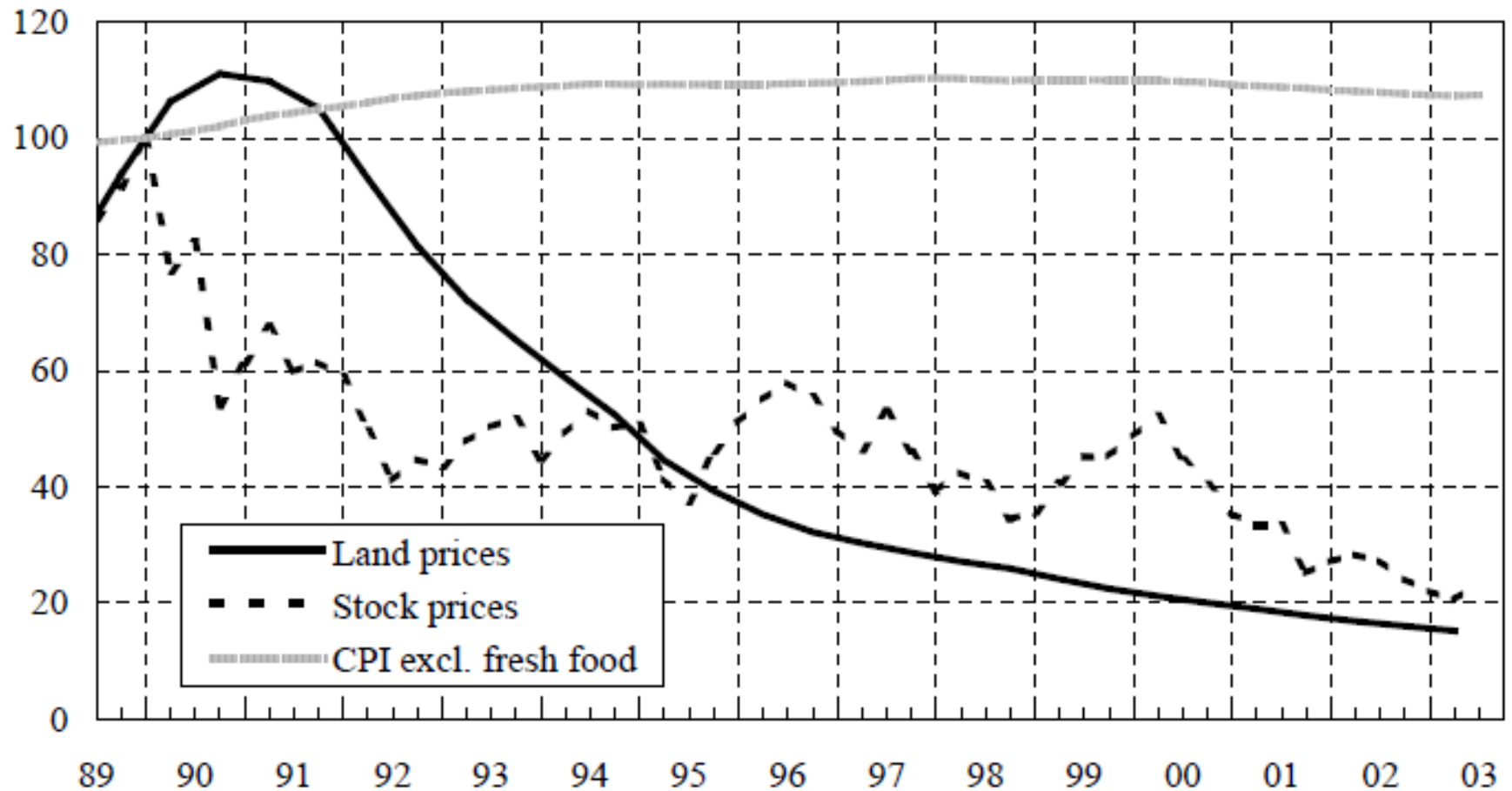
An **asset price bubble** in **real estate** took off...

- ✓ Much of bank lending went into *real estate*, where collateral was increasingly valuable
 - real estate prices skyrocketed
- ✓ For a long time, the Bank of Japan kept interest rates low (due to problem of exchange rate with USD) and the boom went on ...
- ✓ ...in 1990, the bubble reached its peak and burst...
- ✓ in most extreme areas, current real estate values are nowadays around **1%** of their previous levels!

Bubble Burst in Japan

Asset Price Deflation

(1989/IV = 100)



Source: Shiratsuka (2003)

Recession and Deflation in Japan

1. Falling asset prices *eroded bank capital*
 2. but banks were quite good at *hiding* their *losses* (using unrealized gains) initially ...
 3. ... eventually, undercapitalized *banks kept lending to risky borrowers* at good rates so not to recognize losses on their balance sheet: **zombie lending / loan evergreening**
 4. most banks were **bankrupt** but kept alive by the government
- ✓ Japan fell into a severe recession and deflation
 - from which by now it has not recovered yet, despite of very easy monetary policy stance
 - ✓ Government debt has reached 260% GDP... (Germany 60%, Italy 135%, US 125%)
 - optional readings: [Peek and Rosengren \(2005\)](#) and [Caballero, Hoshi and Kashyap \(2008\)](#)

The Great Depression of 1929

- ✓ The **Great Depression of 1929** was another example of a financial crisis followed by debt deflation
- ✓ It has been called the *mother of all financial crises*
- ✓ Its timing varied across nations but in most countries, it started in 1930 (1929 in the U.S.)
- ✓ The world economy did not fully recover until after World War II

The 2007-2009 Financial Crisis

- ✓ Like the Great Depression of 1929 the Financial Crisis of 2007-2009 originated in the US ...
 - ... and then it affected the **entire world**
 - its consequences affected some countries more and for longer (EX.: in *Europe*) than the US

- ✓ We know financial crises have several stages. ...
... this one was no different:
 1. **Credit boom & bust + asset price bubble**
 - Bubble in housing, but also other assets: stocks, MBS...
 2. **Banking crisis (financial crisis becomes recession)**
 3. **Debt Deflation** did **not** happen
 - in fact US GDP and employment recovered relatively quickly

For timeline recommend easy reading (optional): [Brunnermeier \(2009\)](#)

Financial Crisis: the seed

✓ The **seed** (that led to led to the credit/housing boom) was...

1. financial liberalization/innovation

- financial **liberalization**: huge capital flows from RoW to US
- financial **innovation**: securitization of loans

2. but also other key “**macro**” factors

- *globalization* of capital flows made the US net debtor with the RoW
- cheap money from the Fed (very *loose monetary policy*)
 - debated point!

Financial Crisis:

Credit Boom & Financial Innovation

- ✓ All this cheap money needs to go somewhere
- ✓ But interest rates on safe assets (govt bonds) were low, so investors **search for yield** in other products
- ✓ Alternative provided by
 - **Collateralized Debt Obligations (CDOs)**
and
 - **Asset/Mortgage-backed securities (ABS/MBS)**
- ✓ But what are CDOs and ABS/MBS?

Financial Crisis: Securitization

✓ Securitization

- the process of selling, pooling together and repackaging assets into securities (CDOs/ABS/MBS)

✓ CDOs/ABS/MBS

- are securities (bonds) *whose interest payments are backed* by a pool of financial assets (loans, bonds, other types of debt)
- CDO/MBS created in the 1980s, but used massively after 2000

✓ 3 parties involved:

1. the **seller** of the original asset pool (a commercial bank)
2. the **issuer** of the CDO/ABS/MBS (bank-sponsored SPV or GSE)
3. the **investor** (institutional investors: funds, banks...)

✓ With this financial innovation banks go

- from an *originate-to-**hold*** (loans) model
- to an *originate-to-**distribute*** model

Securitization & Banks

- ✓ Why would a bank sell off loans or other assets?
 - so these are *off balance sheet*
 - assets not shown on banks' books
 - they become all part of a “**shadow**” banking system

- ✓ Note that this process
 - *since not shown on books ...* the bank can **lend again!**
 - but it **increases overall leverage** in the economy!

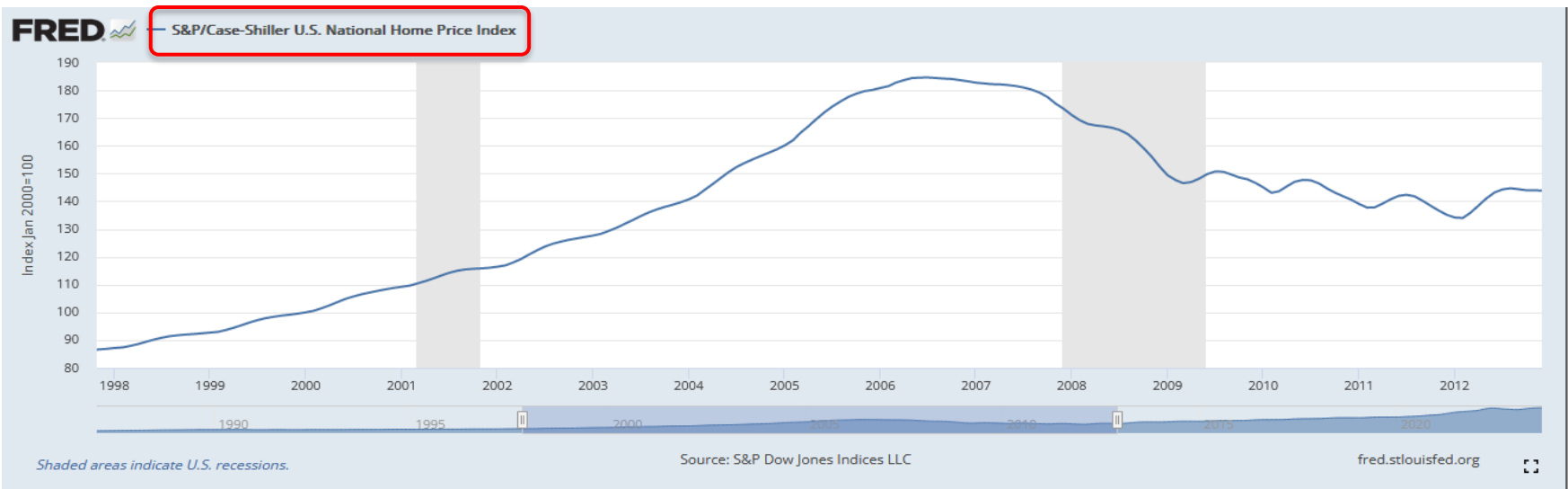
- ✓ Securitization implies “obvious» moral hazard problem
 - if banks take risks off balance sheet, **less incentive to screen and monitor borrowers** (*riskier lending*)
 - moral hazard fueled **subprime lending** *i.e.* loans to individuals with low credit scoring
 - 22% of all US mortgages were subprime in 2006

Financial Crisis: the trigger

✓ The **trigger**?

✓ House prices began to fall in 2006

- most economists failed to predict dramatic effects on the financial system and a major economic crisis ...
 - many said: “just a decrease in demand and a slowdown in growth”
- low house price growth increased *risk-loving investors* defaults
- joined with *securitization of mortgages*... asy info increased!



Where do we stand now? (optional)

✓ Stablecoins

- a major and recent [financial innovation](#)
- [benefits](#)
- [current value](#) of USTD (Tether)

✓ Stablecoins and risks to financial stability

- a new instrument with an old type of risk? Current views:
 - [Krugman](#) (hard view) ... also a bit of intro on stable coins
 - [Portes](#) on multi-issuer stablecoins
 - Regulators ([ECB](#) and [Fed](#))